Dheeraj Raja Kumar

PhD in Applied AI

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Work Experience

- **Scientific Researcher - R2** — Barcelona, Spain
  Centre Tecnològic de Telecomunicacions de Catalunya — *Mar. 2025 - Present*
  - **Overview:** Formulated and implemented AI/ML solutions for specific problems in wireless communication with feedback link. Developed strategies to meet clearly defined performance metrics and compared against conventional ML and mathematical benchmarks.
  - Customized and designed block and convolutional autoencoders models for various use cases based on data availability and system requirements. Worked with Gen AI tools to accelerate coding of large-scale datasets. Performed Exploratory Data Analysis of the communication system with and without AI/ML solutions.
  - Developed deeper understanding of AI/ML strategies to justify the value of model approaches in wireless communication problems. Implemented and maintained efficient data storage and visualization retrieval mechanisms for models using PyTorch.
  - **Relevant fields:** Transformers, Mutual Information Estimation, Reinforcement learning agents.

- **Researcher - R1** — Barcelona, Spain
  Centre Tecnològic de Telecomunicacions de Catalunya — *Sep. 2020 - Feb. 2025*
  - **Overview:** Engaged in research and design of latest AI, high-performance compute algorithms, neural network models, and simulations along with their applications to solve complex problems in wireless communication.
  - Developed simulations in Python and AI/ML framework (PyTorch, TensorFlow, PyG) working on GeForce RTX 3080 GPUs. Fine-tuned machine learning models GNNs, 1D-CNNs, MLPs applied to channel estimation, symbol detection, beamforming.
  - Designed shallow and explainable architectures of neural network models that achieved 10x improvement in performance metric; GNNs were deployed that needed 100x fewer trainable parameters directly affecting computation cost. Multiple peer-reviewed journals and conference papers were published in top-tier IEEE Transactions and Conferences.
  - **Relevant fields:** Time-series forecasting, Anomaly detection, Explainable AI, Graph Neural Network design and processing.

- **Cyber Risk Analyst - Intern** — Hong Kong
  Deloitte Touche Tohmatsu — *Mar. 2018 - Aug. 2018*
  - **Overview:** Internship in the Risk Advisory division focusing on cyber security compliance of a digital banking solution prior to public launch in APAC region.
  - Assisted senior consultants in Pentesting of a client-facing solution in the banking sector.
  - Studied and audited support use cases, UI/UX designs, results documentation to ensure compliance with the polices defined by Hong Kong Monetary Authority (HKMA).
  - Collaborated with teams from both business and technical sides, including users, business owners, engineers, architects, and UI designers to achieve end-to-end project goals; exposed to how digital solutions are integrated into production.

Education

- **CTTC and Universitat Politècnica de Catalunya** — Barcelona, Spain
  PhD in Dept. of Signal Theory and Communications — *Awarded Sep. 2025*
  Title: AI-Driven Multi-Antenna Designs for Next Generations of Wireless Communication Systems
- **Universitat Politècnica de Catalunya** — Barcelona, Spain
  Master's Degree in Applied Telecommunications and Engineering Management — *Feb. 2019 - Jun. 2020*
  Focus: ICT-based Entrepreneurship, Next-generation Wireless Communications
- **City University of Hong Kong** — Hong Kong
  BE (Honours) in Electronic and Communication Engineering; CGPA: 3.4 — *Aug. 2014 - Aug. 2018*

Project Participation (EU and Spain funded)

- **ETSI Industry Specification Group**
  2025-27 Industry Specification Group on Multiple Access Techniques (ISG MAT) — *Jan. 2025 - Present*
  - **Role**: Served as delegate representing CTTC's research interest. ISG members included institutions BBC, Viavi Solutions, EURECOM, Imperial College London, Huawei.
- **Project WAVE-XL**
  2025-28 Spain National Research Project (Ref. PID2024-160457OB-I00) — *Oct. 2025 - Present*
- **COST CA20120 INTERACT**
  2021-2025 COST Action of European Cooperation in Science and Technology — *Sep. 2022 - Sep. 2025*
- **Research Fellow RAUN 2023**
  Regional Academy on the United Nations (RAUN) — *Jun. 2022 - Jan. 2023*

Honors and Awards

- **Shortlisted nominee for ETSI Young Researcher Award 2026**
  Recognizes young researchers whose doctoral dissertation impacts on standardization — *May 2026*
- **2020 Citython Barcelona Winners – EIT Urban Mobility Challenge**
  One of the final winners of city-wide hackathon; Reward of 1.500 EUR. Role: Team lead of 3 members — *2020*
- **Full-Ride Scholar: HKSAR Top Scholarship for International Students**
  Undergraduate Scholarship of 160.000 HKD/year covering tuition and living expenses — *2014 - 2017*

Technical Skills

- **ML & Foundation Models**
  Autoencoders, CNN, GNN, Quantization, Pruning, Efficient Transformers, Distributed Training and Inference
- **Framework & Tools**
  PyTorch, TensorFlow, PyG, Sklearn, Pandas, Nvidia Sionna
- **Programming languages**
  Python, C#, MATLAB